Fab materials prices at a glance

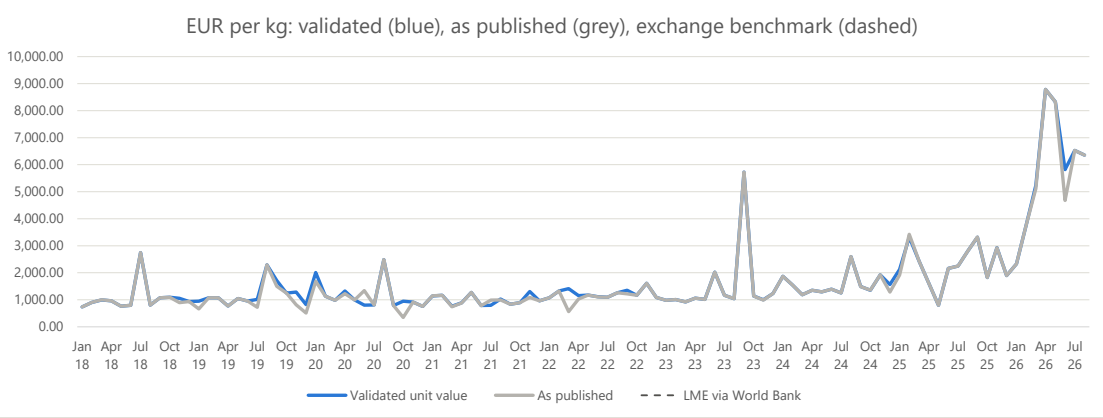
PUBLIC DATA

Validated EU import unit values (EUR/kg, Comext) with exchange benchmarks, reconciliation alerts and nowcasts. Pick a material in the blue cell.

Material GE **Germanium**

Raw materials & metals · Unwrought and powders

LATEST VALIDATED UNIT VALUE	CHANGE vs 12 MONTHS EARLIER	NOWCAST NEXT MONTH (80% INTERVAL)	ALERTS · OUTLIER MONTHS
6,351.07 EUR/kg	**+165%**	**6,569.17**	**2 · 5**
Aug 2026	3-month weighted unit value	from 5,973.01 to 9,572.73 · method ets	source pairs in alert · robust outlier months (QC03)



SOURCE PAIRS FOR THIS MATERIAL (see Reconciliation)

Pair	Comparison	Median gap	Latest gap	Status	Card
R09	EU import unit value vs germanium metal price	-10%	-45%	ALERT	D04
R17	Germany imports from China (CIF) vs China exports to Germany (FOB)	+294%	+314%	ALERT	D08

All materials, by stage of the chain

Stage	ID	Material	Unit value 3m (EUR/kg)	12-month change	Last 24 months	Alerts	Nowcast next month	80% low	80% high	Outlier months
Raw materials & metals	**AL**	Aluminium (unwrought)	3.39	+44%		1	3.34	3.20	3.46	0
Raw materials & metals	**CU**	Copper (refined cathodes)	12.06	+39%		0	12.48	12.32	13.05	0
Raw materials & metals	**GA**	Gallium	1,223.69	+239%		2	1,065.15	722.01	1,363.39	3
Raw materials & metals	**GE**	Germanium	6,304.14	+165%		2	6,569.17	5,973.01	9,572.73	5
Raw materials & metals	**IN**	Indium	377.93	+20%		1	388.99	337.36	475.07	2
Raw materials & metals	**POLYSI**	Polysilicon (Si >= 99.99%)	9.41	-28%		1	9.96	5.50	13.02	8
Raw materials & metals	**SI_MET**	Silicon metal (Si < 99.99%)	1.73	-10%		1	1.75	1.65	1.79	0
Raw materials & metals	**SN**	Tin (unwrought)	46.24	+61%		0	43.04	39.22	45.34	0
Raw materials & metals	**W**	Tungsten (ores and concentrates)	34.99	+12%		0	4.55	1.10	8.84	2
Gases	**HE**	Helium	48.72	+23%		1	25.39	18.94	30.51	10
Gases	**NE**	Neon, krypton and xenon	114.57	-24%		1	181.16	46.07	404.66	4
Chemicals	**H2SO4**	Sulphuric acid	0.34	+129%		0	0.35	0.31	0.47	6
Chemicals	**HF**	Hydrofluoric acid	3.54	+4%		0	3.45	3.08	3.67	2
Chemicals	**NH3**	Ammonia (anhydrous)	0.80	+86%		0	0.67	0.62	0.75	6
Intermediate & finished	**WAFER**	Doped silicon wafers	236.97	-24%		1	252.44	194.19	297.09	1

Sources: Eurostat Comext, World Bank Pink Sheet (CC BY 4.0), USGS, UN Comtrade, ECB. Independent case study based on public data · not affiliated with or endorsed by TechInsights

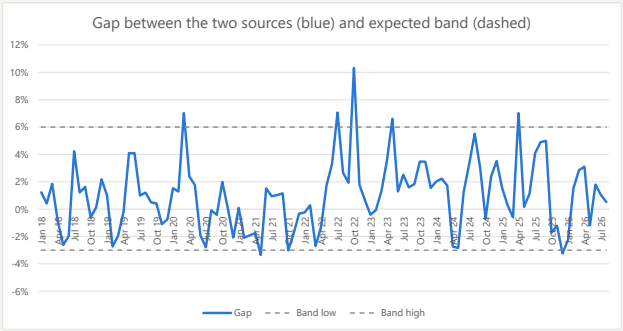
Where sources disagree, and whether the gap is expected

Each pair has an expected band (e.g. CIF import value above the LME price). Red rows = latest gap outside its band. 'card' opens the root-cause card.

PUBLIC DATA

pair_id	material_id	comparison	discrepanc	observations	median_gap	min_gap	max_gap	alerts	first_period	last_period	latest_gap	latest_status	expected_lov	expected_hig	card
R01	CU	EU import unit value (CIF) vs LME cash	D01	104	+1.1%	-3.4%	+10.3%	8	2018-01	2026-08	+0.5%	ok	-3.0%	+6.0%	open D01
R02	SN	EU import unit value (CIF) vs LME cash	D01	104	+0.2%	-21.9%	+30.9%	26	2018-01	2026-08	+3.3%	ok	-6.0%	+8.0%	open D01
R03	AL	EU import unit value (CIF) vs LME cash	D01	104	+8.9%	-9.5%	+33.6%	20	2018-01	2026-08	+18.7%	ok	0.0%	+20.0%	open D01
R04	CU	World Bank vs IMF benchmark		105	+0.0%	-0.5%	+0.5%	0	2018-01	2026-09	+0.0%	ok	-1.0%	+1.0%	
R05	SN	World Bank vs IMF benchmark		105	+0.0%	-1.0%	+0.8%	1	2018-01	2026-09	+0.0%	ok	-1.0%	+1.0%	
R06	AL	World Bank vs IMF benchmark		105	+0.0%	-0.5%	+0.3%	0	2018-01	2026-09	-0.0%	ok	-1.0%	+1.0%	
R07	AL	US market spot vs LME cash	D02	5	+23.5%	+18.0%	+50.8%	1	2021-01	2025-01	+50.8%	ALERT	0.0%	+30.0%	open D02
R08	GA	EU import unit value vs US import unit value	D03	5	-25.1%	-30.9%	-1.5%	1	2021-01	2025-01	-30.9%	ALERT	-30.0%	+30.0%	open D03
R09	GE	EU import unit value vs germanium metal price	D04	5	-9.5%	-44.8%	-1.6%	1	2021-01	2025-01	-44.8%	ALERT	-40.0%	+10.0%	open D04
R10	IN	EU import unit value vs Rotterdam price		5	-11.0%	-18.6%	-4.3%	0	2021-01	2025-01	-11.3%	ok	-25.0%	+10.0%	
R11	SI_MET	EU import unit value vs US market price	D02	5	-33.0%	-52.9%	-18.7%	0	2021-01	2025-01	-22.7%	ok	-60.0%	0.0%	open D02
R12	SI_MET	EU import (CIF) vs China export (FOB) unit value		8	+15.0%	-19.7%	+57.7%	2	2018-01	2025-01	+57.7%	ALERT	0.0%	+40.0%	
R13	NH3	EU import (CIF) vs US Gulf (FOB)		5	+15.4%	-16.7%	+22.4%	1	2021-01	2025-01	+19.2%	ok	-15.0%	+40.0%	
R14	W	EU import per kg of concentrate vs price per kg of WO3 conter	D05	5	-53.0%	-96.0%	-32.0%	0	2021-01	2025-01	-41.6%	not comparable			open D05
R15	POLYSI	EU import vs China export unit value (same CN code)	D06	8	+46.3%	-52.6%	+126.0%	0	2018-01	2025-01	+29.1%	not comparable			open D06
R16	POLYSI	Germany exports to China (FOB) vs China imports from Germa	D07	8	-7.0%	-33.9%	+13.9%	3	2018-01	2025-01	-4.3%	ok	-12.0%	0.0%	open D07
R17	GA+GE+IN	Germany imports from China (CIF) vs China exports to German	D08	8	+293.9%	+75.0%	+2931.8%	8	2018-01	2025-01	+314.5%	ALERT	0.0%	+15.0%	open D08
R18	WAFER	Germany imports from Japan vs Japan exports to Germany (ne	D09	8	+145.4%	+109.4%	+173.9%	8	2018-01	2025-01	+130.0%	ALERT	-10.0%	+10.0%	open D09
R19	MULTI	Same year in MCS2025 vs MCS2026	D10	40	0.0%	-55.7%	+4.6%	7	2021-01	2024-01	0.0%	ok	-5.0%	+5.0%	open D10
R20	NE	Published extra-EU unit value vs validated (QC14) unit value	D11	104	-75.1%	-98.9%	+0.1%	70	2018-01	2026-08	-37.6%	ALERT	-25.0%	+25.0%	open D11
R21	HE	Share of EU helium imports with a confidential partner countr	D12	104	+31.2%	+4.6%	+56.3%	100	2018-01	2026-08	+11.1%	ALERT	0.0%	+10.0%	open D12
R22	POLYSI	EU (changing composition) vs EU27_2020 extra-EU imports in :	D13	25	+24.8%	+12.2%	+58.2%	25	2018-01	2020-01	+39.3%	ALERT	-2.0%	+2.0%	open D13

Pair **R01** EU import unit value (CIF) vs LME cash



Nowcasting the months Comext has not published yet

PUBLIC DATA

Excel's FORECAST.ETS (no seasonality) with an 80% FORECAST.ETS.CONFINT interval, tested on the past; compared with the Python back-test of three methods.

Material **CU** **Copper (refined cathodes)**

EXCEL NOWCAST (FORECAST.ETS)

Month	Nowcast	80% low	80% high	Python (selected)
Sep 2026	**12.48**	12.11	12.84	12.48
Oct 2026	**12.53**	12.08	12.98	12.84

Python selected method: signal

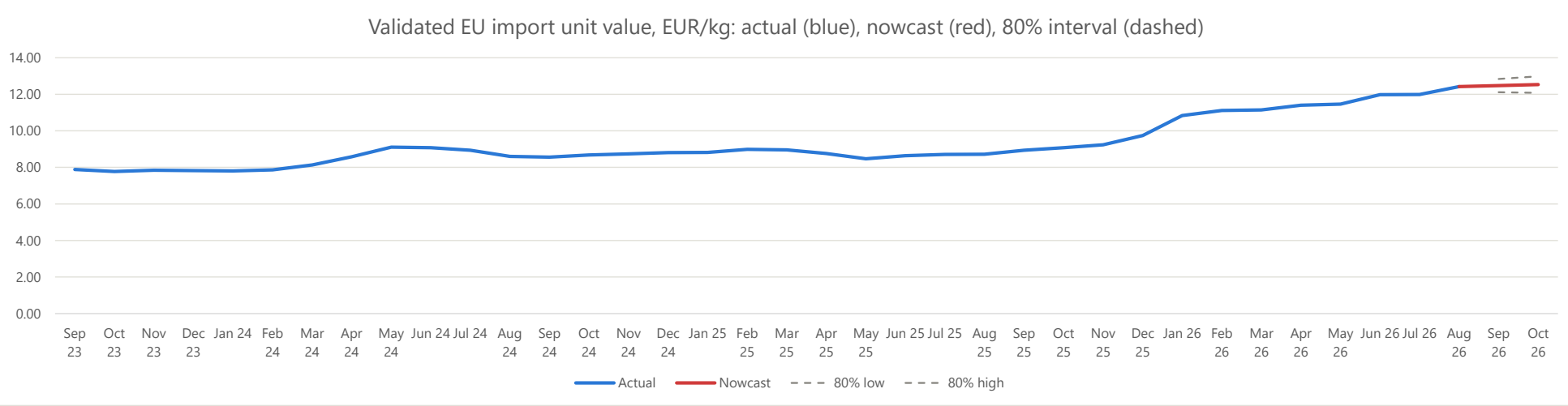
BACK-TEST: IF I HAD FORECAST 6 MONTHS AGO

Month	Actual	Forecast	Abs. error
Mar 2026	11.15	11.17	0.2%
Apr 2026	11.41	11.22	1.7%
May 2026	11.46	11.27	1.8%
Jun 2026	11.98	11.31	5.9%
Jul 2026	11.99	11.36	5.5%
Aug 2026	12.42	11.41	8.8%
Mean absolute % error			**4.0%**

PYTHON BACK-TEST, 24 ROLLING ORIGINS (mean absolute % error by horizon)

Horizon	naive	ets	signal
1	2.4%	2.4%	2.2%
2	4.4%	4.4%	2.4%
3	5.8%	5.8%	2.6%
4	7.1%	7.2%	2.7%
5	8.2%	8.3%	2.9%
6	9.7%	9.5%	3.0%

naive = last value · ets = damped-trend exponential smoothing · signal = regression on the World Bank price (metals only, known one month earlier). Intervals = 10th-90th percentile of past errors.



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Quality checks run on every refresh

14 rules on completeness, units, outliers, nomenclature, confidentiality and freshness (sql/03_checks.sql). Implausible flows are excluded from the validated series.

check_id	check_name	severity	rows_flagged	materials_affected
QC01	Missing month	error	7	2
QC02	Value without quantity	warn	6006	15
QC03	Unit-value outlier	warn	49	11
QC04	Thin trade month	info	24	5
QC05	Nomenclature change	info	4	2
QC08	Inconsistent units (kg vs m3)	warn	47	2
QC09	Partner country not disclosed	info	154	8
QC11	Stale source	info	6	0
QC13	Estimated quantity (Comtrade)	info	56	0
QC14	Implausible partner flow (excluded)	warn	6712	15

Rows flagged per material and check

	QC01	QC02	QC03	QC04	QC05	QC06	QC07	QC08	QC09	QC10	QC11	QC12	QC13	QC14
POLYSI	-	769	8	0	0	0	0	0	1	0	0	0	0	501
SI_MET	0	211	0	0	0	0	0	0	38	0	0	0	0	511
GA	0	699	3	6	0	0	0	0	0	0	0	0	0	98
GE	1	367	5	8	0	0	0	0	0	0	0	0	0	98
IN	0	653	2	1	0	0	0	0	0	0	0	0	0	68
W	6	30	2	8	0	0	0	0	0	0	0	0	0	132
CU	0	201	0	0	0	0	0	0	0	0	0	0	0	371
SN	0	328	0	0	0	0	0	0	0	0	0	0	0	264
AL	0	181	0	0	2	0	0	0	1	0	0	0	0	967
HE	0	397	10	0	0	0	0	5	105	0	0	0	0	903
NE	0	477	4	1	0	0	0	42	2	0	0	0	0	528
HF	0	243	2	0	0	0	0	0	1	0	0	0	0	337
H2SO4	0	185	6	0	0	0	0	0	1	0	0	0	0	1292
NH3	0	189	6	0	0	0	0	0	5	0	0	0	0	539
WAFER	0	1076	1	0	2	0	0	0	0	0	0	0	0	103

Rule names: QC01 Missing month; QC02 Value without quantity; QC03 Robust outlier (median/MAD); QC04 Thin trade; QC05 Nomenclature change; QC06 Code outside validity; QC07 Duplicates; QC08 kg vs m3 units; QC09 Confidential partner; QC10 Withheld reference; QC11 Freshness; QC12 Missing FX; QC13 Estimated quantity; QC14 Implausible flow (excluded)

Errors and the largest unit-value outliers (QC03)

Check	Severity	Material	Period	Detail
QC01	error	GE	2025-04	No extra-EU import record for this month
QC01	error	W	2018-01	No extra-EU import record for this month
QC01	error	W	2018-03	No extra-EU import record for this month
QC01	error	W	2019-01	No extra-EU import record for this month
QC01	error	W	2020-02	No extra-EU import record for this month
QC01	error	W	2020-08	No extra-EU import record for this month
QC01	error	W	2022-10	No extra-EU import record for this month
QC03	warn	HE	2026-08	unit value 79.89 EUR/kg is 213.0% away from its 13-month median
QC03	warn	GE	2023-09	unit value 5725.13 EUR/kg is 390.0% away from its 13-month median
QC03	warn	HE	2026-07	unit value 44.07 EUR/kg is 64.0% away from its 13-month median
QC03	warn	NE	2024-06	unit value 7.25 EUR/kg is -96.0% away from its 13-month median
QC03	warn	GE	2018-07	unit value 2744.29 EUR/kg is 190.0% away from its 13-month median
QC03	warn	POLYSI	2024-07	unit value 7.7 EUR/kg is -75.0% away from its 13-month median
QC03	warn	NE	2022-05	unit value 21.45 EUR/kg is -95.0% away from its 13-month median
QC03	warn	NH3	2023-08	unit value 0.3 EUR/kg is -41.0% away from its 13-month median
QC03	warn	GE	2025-05	unit value 796.91 EUR/kg is -63.0% away from its 13-month median
QC03	warn	POLYSI	2025-10	unit value 3.29 EUR/kg is -72.0% away from its 13-month median
QC03	warn	HE	2026-06	unit value 40.17 EUR/kg is 58.0% away from its 13-month median
QC03	warn	POLYSI	2024-08	unit value 8.33 EUR/kg is -72.0% away from its 13-month median
QC03	warn	POLYSI	2025-12	unit value 2.5 EUR/kg is -72.0% away from its 13-month median
QC03	warn	H2SO4	2019-05	unit value 0.35 EUR/kg is 131.0% away from its 13-month median
QC03	warn	NE	2022-02	unit value 22.29 EUR/kg is -94.0% away from its 13-month median
QC03	warn	GE	2020-08	unit value 2485.41 EUR/kg is 154.0% away from its 13-month median
QC03	warn	H2SO4	2024-02	unit value 0.33 EUR/kg is 121.0% away from its 13-month median
QC03	warn	NH3	2023-05	unit value 0.39 EUR/kg is -38.0% away from its 13-month median
QC03	warn	NH3	2023-11	unit value 0.72 EUR/kg is 63.0% away from its 13-month median
QC03	warn	HE	2026-05	unit value 16.34 EUR/kg is -34.0% away from its 13-month median
QC03	warn	NH3	2023-06	unit value 0.36 EUR/kg is -38.0% away from its 13-month median
QC03	warn	GA	2018-12	unit value 403.46 EUR/kg is 145.0% away from its 13-month median
QC03	warn	GA	2023-02	unit value 169.57 EUR/kg is -59.0% away from its 13-month median

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One standard schema for every source

Each observation, whatever its source, is stored with the same columns; units, currencies and delivery terms are explicit.

Standard schema (data/clean/observations.csv)

period	date	First day of the month (or of the year for annual sources)
frequency	text	M monthly, A annual
material_id	text	Key of the material dictionary
source / series_id	text	Publisher and series code
flow	text	import, export, price, index
reporter / partner	text	Who reports, and the counterpart
valuation	text	CIF, FOB, spot, index, feedstock…
product_code	text	CN8 or HS6 code(s) summed
value_native / currency	number	As published
quantity_kg	number	Net mass in kg (Comext: 100 kg x 100)
quantity_supp / supp_unit	number	Supplementary unit, e.g. m3 for gases
price_native / unit	number	Price in the source's unit
usd_per_eur	number	ECB monthly average (annual mean for annual data)
price_eur_kg / price_usd_kg	number	Standardised price
collected_on	date	Collection date of the snapshot

Material dictionary (seeds/materials.csv)

material_id	material	chain_stage	stage_order	fab_use	standard_unit	purity_or_grade	comext_code	hs6	usgs_series	benchmark_	index_series	notes	plausible_mi	plausible_max	eur_kg
POLYSI	Polysilicon (Si >= 99.99%)	Raw materials	1	Feedstock for wafer ingots	EUR/kg	>= 99.99% Si (solar and electronic grades	28046100	280461			PPI_SEMI	CN8 does not separate solar from electronic grade (9N	1	200	
SI_MET	Silicon metal (Si < 99.99%)	Raw materials	1	Feedstock for polysilicon and silanes	EUR/kg	< 99.99% Si (metallurgical grade)	28046900	280469	USGS_SI_META				0.5	20	
GA	Gallium	Raw materials	1	GaN / GaAs compound semiconductors	EUR/kg	Unwrought and powders	81129289	811292	USGS_GA			China export licensing from 2023-08-01	50	3000	
GE	Germanium	Raw materials	1	SiGe devices; optics and fibre	EUR/kg	Unwrought and powders	81129295	811292	USGS_GE_MET.			China export licensing from 2023-08-01	300	20000	
IN	Indium	Raw materials	1	InP devices; ITO sputtering targets	EUR/kg	Unwrought and powders	81129281	811292	USGS_IN_ROTT				50	2000	
W	Tungsten (ores and concentrates)	Raw materials	1	Contacts and plugs (W CVD via WF6)	EUR/kg	Concentrate (gross weight; WO3 content	26110000	261100	USGS_W_CONC			USGS quotes per metric ton unit (10 kq) of WO3 conter	0.5	100	
CU	Copper (refined cathodes)	Raw materials	1	Interconnects; plating; sputtering targets	EUR/kg	Grade A cathode	74031100	740311	USGS_CU_LME	WB_COPPER,IN	PPI_NONFERR(		2	30	
SN	Tin (unwrought)	Raw materials	1	Solder for packaging	EUR/kg	Unwrought not alloyed	80011000	800110		WB_TIN;IMF_TI	PPI_NONFERR(		8	100	
AL	Aluminium (unwrought)	Raw materials	1	Metallization; sputtering targets	EUR/kg	Unwrought not alloyed	76011000|7601	760110	USGS_AL_SPOT	WB_ALUMINUI		CN 76011000 split into 76011010 + 76011090 on 2023-(	0.8	10	
HE	Helium	Gases	2	Wafer cooling; carrier and purge gas	EUR/kg	Gas (supplementary unit m3)	28042910	280429			PPI_GASES	Quantity reported both in kg and in m3	2	500	
NE	Neon, krypton and xenon	Gases	2	Excimer laser gas mixtures for lithography	EUR/kg	Rare gases excluding helium and argon	28042990	280429			PPI_GASES	Neon cannot be separated from krypton and xenon in (	5	50000	
HF	Hydrofluoric acid	Chemicals	3	Wet etching and cleaning	EUR/kg	All concentrations (electronic grade not s	28111100	281111	USGS_FLUORSI		PPI_INORGANI		0.3	30	
H2SO4	Sulphuric acid	Chemicals	3	Wafer cleaning (SPM / piranha)	EUR/kg	All grades	28070000	280700	USGS_SULFUR		PPI_INORGANI		0.02	5	
NH3	Ammonia (anhydrous)	Chemicals	3	Nitride deposition; cleaning (SC-1)	EUR/kg	Anhydrous	28141000	281410	USGS_AMMON		PPI_INORGANI	USGS quotes FOB Gulf Coast per short ton	0.1	5	
WAFER	Doped silicon wafers	Intermediate &	4	Substrate for chips	EUR/kg	Doped Si discs and wafers	38180010|3818	381800			PPI_SEMI	CN 38180010 split into 38180011 (photovoltaic) + 3818	10	20000	

Customs-code concordance (seeds/code_concordance.csv)

material_id	cn8_code	valid_from	valid_to	cn8_label	change_note
POLYSI	28046100	2017-01		Silicon containing >= 99.99% by weight of silicon	
SI_MET	28046900	2017-01		Silicon containing < 99.99% by weight of silicon	
GA	81129289	2017-01		Unwrought gallium; gallium powders	
GE	81129295	2017-01		Unwrought germanium; germanium powders	
IN	81129281	2017-01		Unwrought indium; indium powders	
W	26110000	2017-01		Tungsten ores and concentrates	
CU	74031100	2017-01		Copper, refined, cathodes and sections of cathodes	
SN	80011000	2017-01		Unwrought tin, not alloyed	
AL	76011000	2017-01	2022-12	Aluminium, not alloyed, unwrought	Split on 2023-01-01
AL	76011010	2023-01		Aluminium slabs, not alloyed, unwrought	New code from 2023-01-01 (part of former 76011000)
AL	76011090	2023-01		Aluminium, not alloyed, unwrought (excl. slabs)	New code from 2023-01-01 (part of former 76011000)
HE	28042910	2017-01		Helium	
NE	28042990	2017-01		Neon, krypton and xenon	
HF	28111100	2017-01		Hydrogen fluoride (hydrofluoric acid)	
H2SO4	28070000	2017-01		Sulphuric acid; oleum	
NH3	28141000	2017-01		Anhydrous ammonia	
WAFER	38180010	2017-01	2025-12	Silicon doped for use in electronics, discs, wafers, cylinders, rods	Split on 2026-01-01
WAFER	38180011	2026-01		Doped silicon wafers <= 200 micrometres, rectangular (photovoltaic	New code from 2026-01-01 (part of former 38180010)
WAFER	38180019	2026-01		Silicon doped for use in electronics (excl. photovoltaic wafers)	New code from 2026-01-01 (part of former 38180010)

Discrepancy log: finding, root cause, evidence, correction, prevention

PUBLIC DATA

Generated from seeds/discrepancy_register.csv with evidence recomputed from the data at each pipeline run.

discrepan	title	materials	root_cause	correction	prevention	status	evidence
D01	EU import unit values sit above the LME cash price	CU; SN; AL	Valuation basis. Comext imports are valued CIF at the EU border (freight, insurance and the regional physical premium are included). LME cash is an exchange price for metal in LME warehouses. Aluminium carries the largest premium (EU duty-unpaid premium). Contracts are often priced on the previous	Compare against the benchmark with the expected premium band, not against zero. Use the 3-month weighted unit value in fast markets.	Expected band per pair in seeds/reconciliation_pairs.csv; an alert is raised only when the gap leaves the band.	Explained	Copper: median gap +1% (band -3% to +6%), 8 alert months out of 104; Tin: median gap +0% (band -6% to +8%), 26 alert months out of 104; Aluminium: median gap +9% (band +0% to +20%), 20 alert months out of 104. Control: World Bank vs IMF copper median gap +0%, so the benchmarks agree and the gap is in the EU valuation.
D02	US market prices are well above EU and LME prices	AL; SI_MET	Market scope. USGS prices for aluminium and silicon metal are US market prices. They include US-specific trade measures: the Section 232 tariff on aluminium (25% from 2025-03-12, 50% from 2025-06-04) and anti-dumping / countervailing duty orders on silicon metal from several origins.	Never mix a US market price with an EU border or LME price in one series. Keep the US series as a separate, labelled market.	'valuation' and 'reporter' (market) are mandatory columns of the standard schema; pairs across markets carry a wide band.	Explained	US aluminium spot vs LME: gap +18% to +51%, latest year +51%. EU silicon metal import unit value vs US market price: median -33%.
D03	Gallium 'price' fell 56% between two USGS editions for the same year	GA	Definition change, not a market move. MCS2025 published a high-purity gallium price; MCS2026 publishes the average unit value of US imports for the same years.	Treat the two editions as two different series; reconcile Comext against the MCS2026 import unit value (same concept).	Store the full definition text with every series and edition; a changed definition text opens a new series instead of overwriting the old one.	Explained	Same years, two editions: 2021: 625 -> 277 USD/kg (-56%); 2022: 560 -> 432 USD/kg (-23%); 2023: 450 -> 365 USD/kg (-19%); 2024: 500 -> 439 USD/kg (-12%). After aligning definitions, EU vs US import unit value median gap is -25%.
D04	EU germanium unit value lags the quoted metal price	GE	Timing and mix. Customs values follow contract prices agreed earlier, while the quoted price reacts at once; EU imports also mix powders and lower purities. USGS 2025 is an estimate made before year-end.	Read the gap with a lag; flag the latest USGS year as provisional.	QC10 / provisional flag on estimated values; nowcast uses the validated EU series only.	Explained	EU import unit value vs USGS metal price (USD/kg): 2023: 1,260 vs 1,392; 2024: 1,593 vs 1,991; 2025: 2,265 vs 4,100.
D05	Tungsten concentrate cannot be compared per kg	W	Content basis. USGS quotes per metric ton unit of WO3 content (10 kg of WO3); Comext records the gross weight of ores and concentrates of very different grades. The ratio of the two implies a WO3 content that swings from a few percent to more than half.	Do not use the EU unit value as a tungsten price; keep it for volumes only.	comparison_basis = 'content_basis' in the data standard: such pairs never raise a price alert.	Not comparable	EU unit value per kg of concentrate divided by the USGS price per kg of WO3 ranges from 4% to 68% (implied WO3 content), far outside a physical 50-75% range for a single concentrate grade.
D06	One customs code holds two different products (polysilicon)	POLYSI	Product mix. CN 28046100 covers every silicon >= 99.99%, from solar-grade to electronic-grade. EU imports from the United States are priced like electronic grade while imports from China are priced like solar grade; the average moves with the origin mix, not with prices.	Track an origin-stratified unit value (US-origin as an electronic-grade proxy) next to the all-origin average.	Mix check: compare the all-origin unit value with a fixed-origin-weight version every month.	Explained	EU import unit value by origin (EUR/kg, tonnes): 2022 CN: 35.6 EUR/kg on 237 t; 2022 US: 32.2 EUR/kg on 4,775 t; 2025 CN: 2.4 EUR/kg on 7,272 t; 2025 US: 41.9 EUR/kg on 2,190 t.
D07	Germany's polysilicon exports to China vs China's imports from Germany	POLYSI	Mirror statistics. Germany values exports FOB, China values imports CIF, so China's figure should be a few percent higher. Larger gaps come from timing (shipments recorded in different years) during the 2023 price collapse.	Use a CIF/FOB band for mirror pairs; investigate only gaps outside it.	Mirror check with an expected band, run on every annual refresh.	Partly explained	Value gap (DE exports FOB / CN imports CIF - 1): median -7%, range -34% to +14%, 3 alert years.
D08	Germany imports far more Chinese gallium/germanium/indium than China says it exports to Germany	GA; GE; IN	Partner definition. The EU records imports by country of ORIGIN; China records exports by country of DESTINATION. Chinese-origin metal bought through traders and shipped from third countries counts as 'China' in Germany's data but not as 'Germany' in China's data.	Do not reconcile small high-value metals on bilateral mirror flows; reconcile on world totals and unit values.	Partner definition documented per source in DATA_STANDARDS.md.	Explained	Germany's reported value is 1.7x to 30.3x China's mirror figure (median 3.9x).
D09	Germany reports twice the wafer weight that Japan reports	WAFER	Open. Values agree within the CIF/FOB band but the net weights do not. Hypotheses: different quantity basis (carriers or packaging counted in one source), or weight estimated by one declarant.	Reconcile wafers on value, not on kg; do not publish a per-kg wafer price from mirror data.	Value gap and kg gap are checked separately; escalate kg gaps above 25% to the source (Destatis / Japan Customs).	Open	Net kg reported by Germany vs Japan: 2.1x to 2.7x, while the value gap stays between +15% and +53%.
D10	Latest-year USGS values are revised in the next edition	ALL	Revisions. The latest year of each edition is an estimate and is revised one year later (several percent for silicon, sulfur, germanium and indium).	Always load the latest edition; flag the last year as provisional.	Keep every edition as a dated snapshot and recompute the revision table at each refresh.	Explained	2024 values, MCS2026 vs MCS2025: SULFUR -7.2%; SI_METAL -5.4%; GE_METAL -5.2%; FLUORSPAR_ACID -1.3%; CU_LME -1.3%; AL_SPOT -0.4%; W_CONC +0.8%; IN_ROTTERDAM +3.7%
D11	Published neon/krypton/xenon unit value is polluted by implausible flows	NE	Misclassification or wrong quantity in customs declarations: hundreds of tonnes declared under CN 28042990 at about EUR 2/kg, a value impossible for neon, krypton or xenon.	Exclude partner flows outside the plausible price range (QC14) and publish the validated series.	Plausible price range per material in seeds/materials.csv; QC14 runs on every refresh.	Corrected	Largest excluded flows: 2024 CH: 2,473 t at 1.36 EUR/kg; 2022 CH: 814 t at 1.77 EUR/kg; 2021 CH: 769 t at 0.90 EUR/kg. Published vs validated unit value: median gap -75%, 70 months outside +/-25%.
D12	A third of EU helium imports have a confidential origin	HE	Confidentiality. Comext books the trade of some declarants under 'secret' or 'not specified' partner codes.	Report 'Confidential' as its own origin instead of spreading it over named countries.	QC09 alerts when the confidential share is above 10%.	Explained	Confidential partner share of EU helium imports, 2018-latest: 31% of value; 100 months above 10%.
D13	'EU' totals before 2020 do not match EU27 totals	POLYSI; CU	Coverage. The Comext 'EU' reporter includes the United Kingdom until January 2020; flows between the UK and non-EU countries are counted as extra-EU before Brexit and UK-EU flows as intra-EU.	Use the fixed composition EU27_2020 for the whole history.	Reporter code fixed in the data standard; composition check kept in the pipeline.	Corrected	Polysilicon extra-EU imports, 'EU' vs EU27_2020 before February 2020: median gap +25% (range +12% to +58%).
D14	Helium weight and volume do not match physics	HE	Units. The kg reported per m3 is about 60% above the density of helium gas at standard conditions: m3 may be declared at other conditions, or part of the declared mass may not be helium.	Use kg as the standard unit; keep m3 only as a check.	QC08 compares the two quantities every month.	Partly explained	Median implied density of EU helium imports: 0.286 kg/m3 vs 0.179 kg/m3 for helium at 0 C and 1 atm (+60%).

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